Controller-Guided Search for BiomedSQL

Draft

Abstract

BiomedSQL asks a model to turn a biomedical question into SQL whose result set matches a gold execution, including thresholds the question never states. We place two roles, a controller and a database agent, between the question and a local SQLite copy of the knowledge base. They share one model and one dialogue. The controller writes a search brief; the database agent scores how well it understands the brief; the controller scores whether the brief is specific enough. Search starts once both scores reach 8, or after six turns. The database agent then runs a short ReAct loop and returns the rows. On the 546-question development sample, Qwen3.8-27B reaches 69.2% execution accuracy and 71.0% Jaccard overlap, with 0.2% of questions producing no executable SQL. The prompts were revised on this same sample, and the database is SQLite rather than BigQuery.

1 Introduction

BiomedSQL is a text-to-SQL benchmark over gene–disease associations, summary-data Mendelian randomization, and drug-approval records [Koretsky et al., 2026]. A correct query often has to supply a convention the question leaves implicit: genome-wide significance at $p < 5 \times 10^{-8}$, an SMR threshold of $p_{\mathrm{SMR}} < 2.95 \times 10^{-6}$, a HEIDI filter, a disease code, or a phase-4 approval. The public development sample has 546 questions from 38 templates. Execution accuracy (EX) requires the returned row set to match the gold result [Yu et al., 2018]. Jaccard (JAC) scores partial overlap. SER is the share of questions with no executable SQL. The published best system, BMSQL-GPT with o3-mini, reaches 62.6% EX, 69.2% JAC, and 2.6% SER. Gemini-3-Pro under a baseline prompt reaches 58.1% EX. We test one open model, two prompts, and a dialogue that has to name the missing convention before any SQL is issued.

2 Method

Both roles call the same Qwen3.8-27B endpoint. The controller sees the question and writes a search brief: table, p-value column, threshold, disease or drug fields, and whether the answer is a row set. The database agent scores its understanding of that brief from 0 to 10 and names any missing constraint. The controller scores satisfaction from 0 to 10. The loop stops when both scores are at least 8, or at six turns. On this run both scores cleared the bar on 545 of 546 questions, after 1.10 turns on average, and 498 questions finished in one turn.

The database agent then searches with a ReAct loop [Yao et al., 2023]: a short thought, one `SELECT`, the rows, and a decision to stop or continue. The cap is six rounds and five successful selects. The mean was 2.30 rounds (median 1); 76 questions used all six. A last prompt turns the rows into a short answer. Scoring uses the rows. BigQuery paths are rewritten to local table names, booleans to 0/1, and double-quoted literals to SQLite strings. A query keeps its own `LIMIT` up to 5,000 rows and otherwise receives `LIMIT 100`. The five tables used by the gold queries are loaded: the Alzheimer and Parkinson gene catalogs, the SMR gene table, drug-target annotations, and indications.

Conventions in the prompt. A 38-question pass, one item per template, scored 57.9% EX. GWAS questions were run as SMR tests, and the reverse. Drug names were matched on one column.

`COUNT(*)` replaced the row set. Approval questions joined annotations and returned drugs the indications table does not mark as approved. The revised prompt states the choice. GWAS wording uses p or $p_{\mathrm{GWAS}} < 5 \times 10^{-8}$ on the disease catalogs. An explicit SMR question uses $p_{\mathrm{SMR,multi}} < 2.95 \times 10^{-6}$. A functional question adds $p_{\mathrm{HEIDI}} > 0.01$. If the question names neither test, the agent tries the GWAS reading and then the SMR reading. Drug lookup uses name, synonyms, and trade names. "Approved to treat" reads only the indications table, at maximum phase 4. Listings keep `LIMIT 1000`, return the UUID, and keep the rows. Before stopping, the agent checks that every constraint in the brief is still in the SQL.

3 Results

Table 1 places this run next to systems from Koretsky et al. [2026]. We match 378 of 546 result sets (69.2% EX), with JAC 71.0% and SER 0.2%. EX is 6.6 points above BMSQL-GPT-o3-mini and 11.1 points above Gemini-3-Pro. JAC is 71.0%, against 69.2% for that BMSQL system. ReAct-o3-mini, which interleaves reasoning and SQL without this brief, is at 56.2% EX.

Table 1: Development sample (n=546). Published rows are from Koretsky et al. [2026] on BigQuery. Ours is Qwen3.8-27B on SQLite. RQR and SR were not computed.

System	EX	JAC	SER
Expert (human, n=20)	90.0	90.0	0.0
Ours (Qwen3.8-27B)	**69.2**	**71.0**	**0.2**
BMSQL-GPT-o3-mini	62.6	69.2	2.6
DAIL-SQL-o3-mini	61.2	63.6	0.0
BMSQL-GPT-4o	60.4	67.2	4.9
Gemini-3-Pro, baseline prompt	58.1	62.4	0.0
ReAct-o3-mini	56.2	64.8	0.0
o3-mini, baseline prompt	53.5	60.4	0.2

The dialogue rarely negotiates: 498 of 546 questions clear both scores on the first turn, so the gain is in what the brief must contain. Questions that used all six SQL rounds were those whose first statement dropped a threshold, a UUID, or a disease filter. One question was closed as a miss. Its follow-up tried to split a comma-separated synonym field with a recursive query and did not return; the first statement had already omitted the UUID, so the recorded EX is 0.

4 Limitations

The official evaluation runs on BigQuery. Ours runs on SQLite, with identifier rewriting and a row cap. Gold queries for these 546 questions use only the five loaded tables and no joins, and a template check of that rewrite scores EX 1.0. An engine difference can still change an edge case. We report EX, JAC, and SER only; reasoning score, semantic score, and token counts are absent. The prompt was revised after the 38-question smoke and a second look at 16 misses from this same sample, and the full sample was then scored with that prompt. Read 69.2% as the score of the revised system on the sample used to revise it.

References

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